PLAYTEST EDITION · BUILD 0.17

FOR FACILITATORS & INSTRUCTORS

COMPANION TO THE FIELD MANUAL

# COGNITIVE·TERRAIN

## PRACTITIONER'S MANUAL · RUN IT · BUILD IT · KNOW WHY




### RUN IT WITH A GROUP

Session plans from a lunch break to a multi-organisation exercise, and how to lead the debrief.

### BUILD YOUR OWN CAMPAIGN

The campaign and technology editors, balance checks and sharing.

### WHY IT WORKS THIS WAY

Design notes: what each mechanic models, and how it was arrived at.

EDUCATIONAL WARGAME · TURN-BASED
PLAYERS NEED ONLY THE FIELD MANUAL

CT · PM · 2026

FOR WHOEVER RUNS THE ROOM

# BEFORE YOU BRIEF

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### WHO THIS IS FOR

Instructors, exercise designers and facilitators who run Cognitive Terrain for other people, build campaigns for their own audience, or need to explain why the game models cognitive warfare the way it does.

Players need only the **Field Manual**, which is also built into the game as the Guide (**?**). This manual assumes you have read it or played a campaign.

### WHAT PLAYERS SHOULD LEAVE WITH

The target of cognitive warfare is the quality of the other side's decisions, not territory or attention.

Reach is not effect: a viral post on the wrong area changes nothing that matters.

Attack and defence happen at once, and open and closed societies are vulnerable in different ways.

### FIRST TIME RUNNING IT?

Play **Typhoon Ines** on **Standard** once yourself, end to end, before you run it for anyone else. It takes about forty minutes.

FOUR FORMATS · SUGGESTED TIMINGS

# RUNNING A SESSION

*The game runs offline on any computer with a browser, so the format is set by your time and your room, not by the technology. Timings are suggestions; adjust them to your group.*

## SOLO

ONE PLAYER · 30–45 MIN

1. Send the link or the HTML file with the Field Manual.
2. Ask for **Typhoon Ines** on **Recruit** with the tutorial first.
3. Ask them to bring a screenshot of the debrief and one turning point from the replay.

## CLASSROOM

GROUPS OF 2–3 · 50–90 MIN

1. 10 min: the big idea (Field Manual page 3) on the projector.
2. 30–60 min: each group plays one campaign on one screen, one driver, others advising.
3. 15–20 min: debrief as a room (page 5 of this manual).

## PAIRED SEMINAR

BLUE vs RED · 60–90 MIN

1. Pair students; one plays Blue, one Red, on two computers in the same room.
2. They swap turn codes by chat (page 6). Sixteen weeks takes about an hour.
3. Debrief each pair, then compare how Red was played across pairs.

## DISTRIBUTED EXERCISE

SEVERAL ORGANISATIONS · DAYS

1. Build or choose a campaign and send the file to every cell first (page 9). Add a White Cell to steer it (page 7).
2. Cells play one week per day or per session, codes by email.
3. Collect debriefs and replays for a joint hot-wash.

### WHAT TO DECIDE BEFOREHAND

**Plain or military terms.** Plain words suit mixed audiences; military terms (OODA, DQ, Seed Channels) suit staff courses. Players switch in the top bar.

**Campaign and difficulty.** See page 4.

**Week in review on or off.** On for learning; off for a faster, harder exercise.

### KEEP IT MOVING

Set Red's turn speed to **FAST** once players know what Red's moves look like, and result pop-ups to **KEY ONLY**. Both are in Settings and change nothing in the rules.

EIGHT CAMPAIGNS · THREE DIFFICULTIES

# CHOOSING A CAMPAIGN

*Every campaign uses the same rules. What changes is the setting, the opponent, the twist at the start and which areas the four crises test. Pick by what you want the room to argue about.*

| CAMPAIGN | VS | TWIST AT THE START | CRISES TEST | GOOD FOR |
|---|---|---|---|---|
| **Typhoon Ines** | Olvana | None | Notice · Interpret · Act · Trust | First games; the core loop |
| **The Valmora Vote** | Olvana | Both TRUST areas weakened | Interpret · Notice · Decide · Trust | Elections, polarisation |
| **Collision at Varrin Reef** | Olvana | Both officials' entry points weakened | Notice · Act · Act · Decide | Crisis diplomacy, competing videos |
| **Fever in Port Sabre** | Olvana | Rumours spread faster for both | Interpret · Trust · Act · Decide | Infodemics, public health |
| **Winter Grid** | Donovia | Blue's FOLLOW-THROUGH strained; 4 dormant narratives | Notice · Decide · Act · Trust | Energy coercion |
| **The Lake Referendum** | Donovia | Donovia already in Blue's INTERPRET; TRUST weakened | Interpret · Notice · Decide · Trust | Referendums, planted premises |
| **The Silent Survey Ship** | Donovia | Both DECIDE areas strained | Notice · Act · Interpret · Decide | Ambiguity, alliance choices |
| **The Vell Valley** | Donovia | 5 dormant narratives; Blue's NOTICE and TRUST strained | Interpret · Act · Trust · Decide | Annexation and its cover story |

## DIFFICULTY

| LEVEL | WHAT CHANGES |
|---|---|
| **Recruit** | Red is slower, makes more mistakes and earns 2 fewer points a week. For first games and short sessions. |
| **Standard** | A steady, sensible Red. Campaigns are balanced at this level. |
| **Veteran** | Red plans two moves ahead. No extra budget, no hidden information. For experienced players and competitions. |

### MATCH THE OPPONENT TO THE LESSON

**Olvana** teaches patience against nerve: a closed state that works one area at a time from inside.

**Donovia** teaches that the aim can be doubt and inaction rather than persuasion: floods, planted premises, shows of force.

Field Manual pages 24 and 25 set out each one's rules and sources.

### SAME MAP, FAIR COMPARISON

Groups playing the same campaign get different maps unless they start from the same seed. For a like-for-like comparison, run it as a two-player game from one Blue code (page 6).

THREE VERDICTS · THE REPLAY · QUESTIONS

# LEADING THE DEBRIEF

*The game ends with a neutral debrief and a replay of both minds week by week. Most of the learning happens here, so leave time for it.*

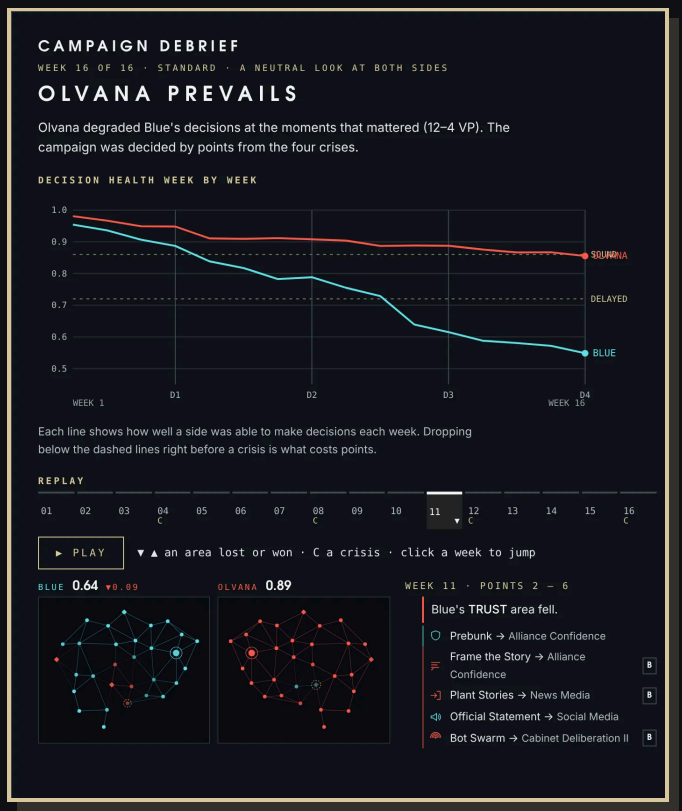



**FIG. 1** The debrief: crisis points, health week by week, three verdicts, turning points and lessons.

## THE THREE VERDICTS

| VERDICT | MEASURES |
|---|---|
| **Who decided better** | Decision Health and crisis results across the campaign. |
| **Who adapted more** | How many of the eight capabilities each side used, and how many adaptive or protective moves. |
| **Who got more for less** | Influence moved per point spent attacking. |

A side can win on points and lose two of the three verdicts. That gap is often the best conversation in the room.




**FIG. 2** The replay: step through every week, with both minds and that week's moves.

## QUESTIONS THAT WORK

- Which crisis did you lose, and when did you first know it was coming?
- Find a move with a great spread score that changed nothing. Why?
- When did defending your own mind matter more than attacking theirs?
- What did Red seem to be trying to do? What told you?
- Which of Blue's advantages as an open society did you use, and which did you lose?
- What would you tell a decision-maker in the real version of this crisis?

TWO PLAYERS · NO SERVER · NO SHARED NETWORK

# ACROSS ORGANISATIONS

*Two-player games need nothing but a way to send a line of text. That makes the game usable between cells, units or agencies that cannot share a network or install software.*

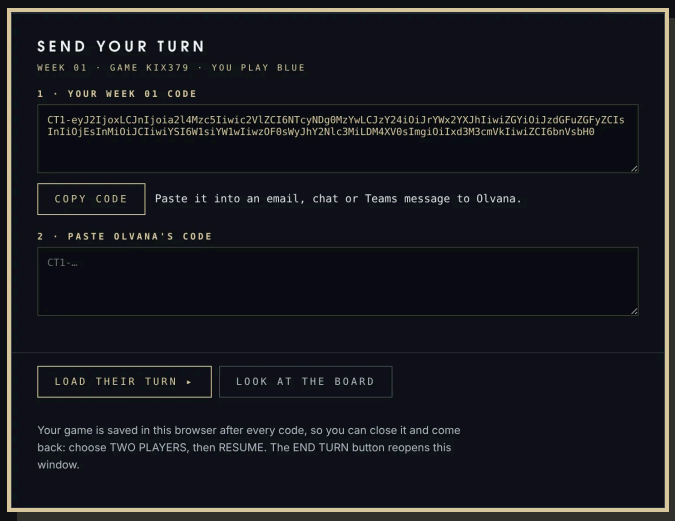



**FIG. 3** A turn code with a COPY button, and a box for the reply.

### WHAT A CODE CARRIES

The game and its map seed, the campaign, the week, the side and that side's moves. Each copy replays the other's moves and checks the board matches; if it ever does not, the log says so. Codes contain no personal data.

### CHECKLIST

1. **Same version.** Codes work only between builds with the same minor version (0.17.x with 0.17.x). Send everyone the same download.
2. **Same campaign.** For a custom campaign, every Red player imports the file before the first code (page 9). The code checks it.
3. **Assign sides.** Blue starts the game; Red joins with Blue's first code. The campaign decides whether Red is Olvana or Donovia. To steer it yourself, start a structured session and join as the White Cell (page 7).
4. **Set a rhythm.** One week per session or per day keeps a sixteen-week campaign moving.
5. **Collect the debriefs.** Each side sees the same debrief at the end; Red sends one final code so Blue's game ends too (in a structured session, the White Cell sends it).

### ANY CHANNEL WORKS

Email, chat, Teams, a text file on removable media. Extra text around a pasted code is ignored, so players can add a note.

Each player's game saves after every code and appears under **RESUME**, so a game can pause for days.

STRUCTURED SESSIONS · A THIRD PLAYER WHO STEERS

# THE WHITE CELL

*In a structured session a facilitator joins as the **White Cell**. It sees both minds, and once a week, after Red and before Blue, it can adjust the game: run a story, change a budget, move a crisis, strike a move. It is the game's version of the control cell in a staff exercise.*

**BLUE** Plays the week Code to Red → **RED** Plays the week Code to White Cell → **WHITE CELL** Watches both, then steers Code to Blue → **BLUE** Reads the changes, plays Next week begins

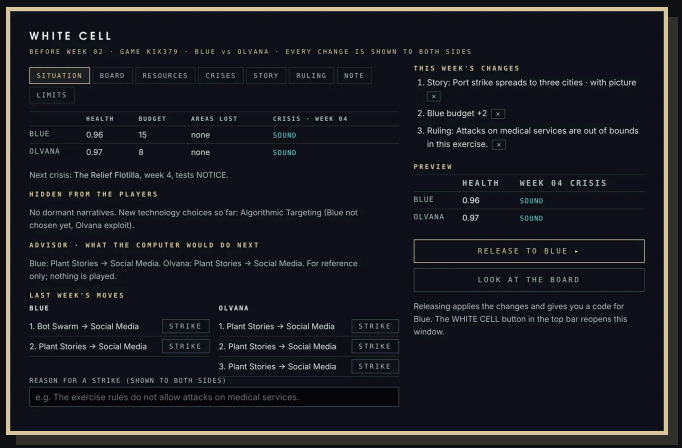



**FIG. 4** The White Cell console before week 2: both sides' health, budget and next crisis; what is hidden from the players; last week's moves with STRIKE buttons; this week's changes and their effect on the right.

### ONLY WHEN YOU CHOOSE IT

Ordinary two-player games are unchanged. The White Cell exists only when Blue starts a **NEW STRUCTURED SESSION · WITH WHITE CELL**.

## SETTING ONE UP

1. **Blue starts it.** TWO PLAYERS → NEW STRUCTURED SESSION. Blue plays week 1 and sends the code to Red.
2. **Red joins** with Blue's code, as in any two-player game, and sends its code to the White Cell, not to Blue.
3. **The White Cell joins** with Red's first code: TWO PLAYERS → JOIN AS WHITE CELL. Both week 1 turns replay and the console opens.
4. **Release.** Queue changes, or none, and press RELEASE TO BLUE. Send the code to Blue. Week 2 begins.

### WHAT THE WHITE CELL SEES

Both minds with no fog, both budgets, every move as it replays, Red's secret technology choices and Blue's dormant narratives. It plays no moves of its own.

The **WHITE CELL** button in the top bar reopens the console; **LOOK AT THE BOARD** closes it to study the map.

### WHAT THE PLAYERS SEE

Nothing is secret. Before Blue's turn, both sides get a White Cell notice with every story, ruling and change (page 8). It is also in the log, the debrief and the replay.

THE CONSOLE · ONCE A WEEK · EVERY CHANGE VISIBLE

# STEERING THE EXERCISE

*Eight tabs, one queue, almost all of it buttons. Each change is listed on the right as you add it, with a preview of both sides' health and the next crisis. Remove any item before you release; nothing happens until you do.*

**SITUATION**
Health, budgets, the next crisis, hidden facts, and what the computer would play next. STRIKE a move here.

**BOARD**
Pick a node on the map, then one click to shift control, set or clear a status, or plant a dormant narrative.

**RESOURCES**
One click to add or take budget points, or give either side a move more or fewer this week.

**CRISES**
Move, retarget or double a coming crisis, add a new one, or schedule a new technology.

**STORY**
A breaking news item, with an optional picture, that both sides read before Blue's turn.

**RULING**
An adjudication the players must follow, with your reason.

**NOTE**
A teaching note kept out of play; it appears in the debrief and replay.

**LIMITS**
Your own caps: largest shift and budget change; switch moves, crises or strikes off.

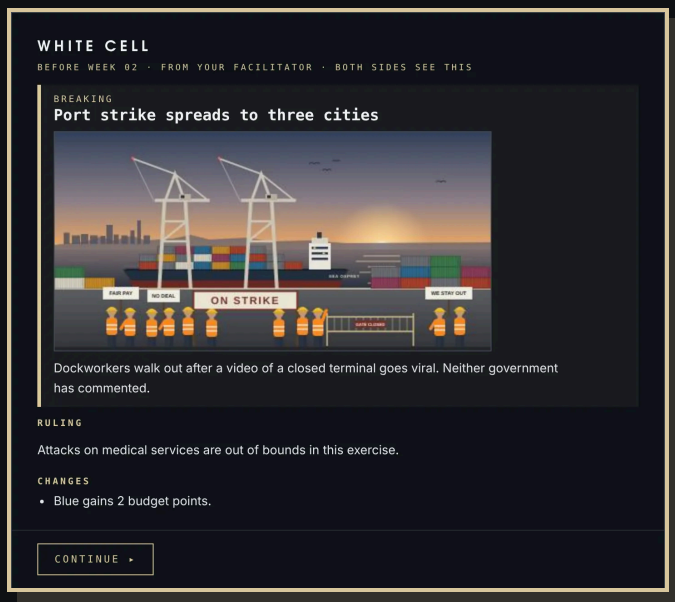



**FIG. 5** What both players see before Blue's week 2: the story and its picture, the ruling and the changes, in plain words.

## RULES THE GAME KEEPS FOR YOU

Control stays between −100 and 100; budgets between 0 and the cap; statuses last at most four weeks. Crises cannot move into the past or onto another crisis. Dormant narratives live only in Blue's mind.

A **strike** rewrites a week already played: every copy rebuilds itself without that move, and both sides are told why.

## INJECTS THAT TEACH

| TO SHOW | TRY |
|---|---|
| Events outrun preparation | Bring the next crisis forward a week, or make it test the area a side has neglected. |
| An outside shock | A story plus a small control shift on the matching node, e.g. a viral clip and a nudge on Social Media. |
| Resource pressure | Cut a side's budget by 2 before a crisis and watch what it gives up. |
| Rules of engagement | A ruling, then strike any move that breaks it. |
| A losing side recovers | One extra move for a week; ask in the debrief what it did with it. |

**POINT AND CLICK**

Click any node on the map and the console opens on it, ready for a shift or a status. Drop, paste or browse for a picture to run with a story: it is shrunk to fit inside the code, so it travels by email or chat like any other code.

CAMPAIGN EDITOR · SHARE · IMPORT

# MAKE A CAMPAIGN

*Set the scene for your own audience, place the crises and new technologies on the calendar, tune how strong each side starts, then test it, check its balance and share it. No code involved.*

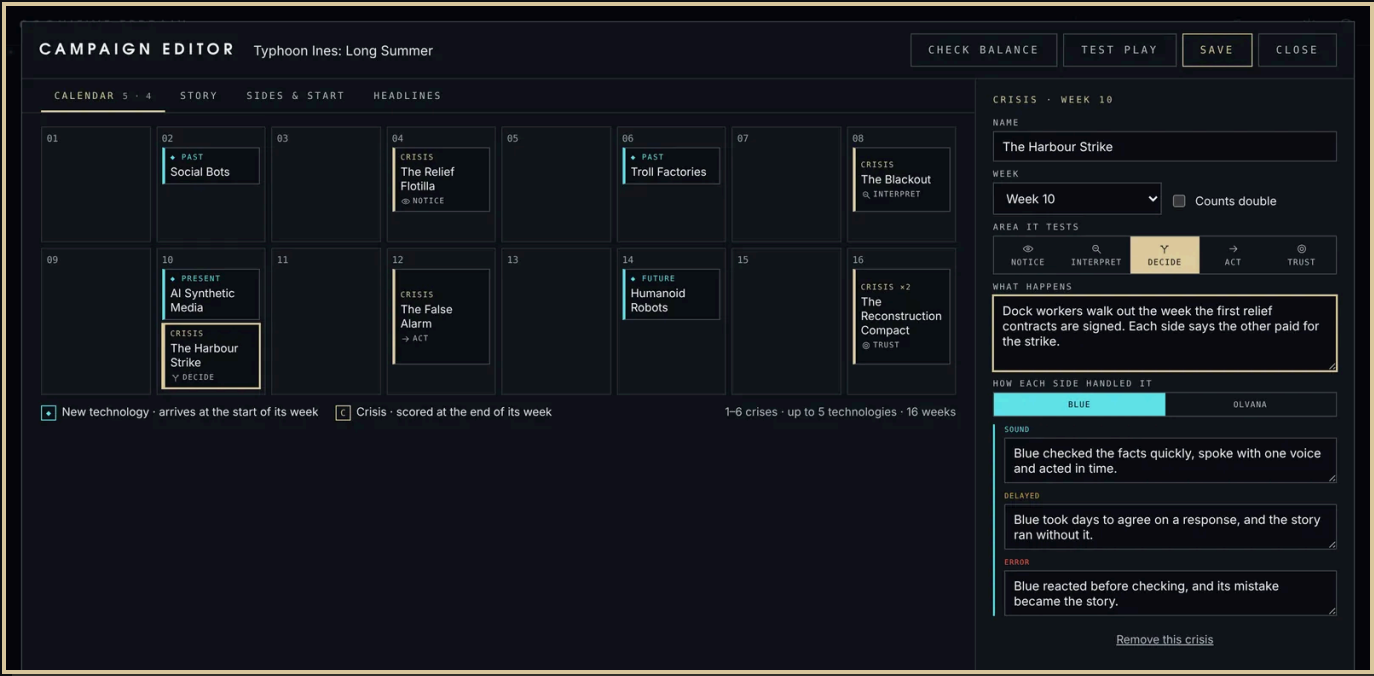



**FIG. 6** The campaign editor, editing a copy of Typhoon Ines with a new crisis in week 10. Click a crisis or technology on the calendar to edit it on the right; + adds one in that week.

## MAKE ONE

1. On the start screen press **CAMPAIGNS** (or ⚙ → **CAMPAIGN EDITOR**).
2. **+ NEW CAMPAIGN**, or **COPY & EDIT** a built-in one. Copying is the fastest way to a balanced start.
3. Choose the opponent (Olvana or Donovia), write the setting and the headlines' places and outlets, and set any twist at the start.
4. Place 1 to 6 crises and up to 5 new technologies (page 10).
5. **CHECK BALANCE**, then **TEST PLAY**.

### SHARING

Save a **.campaign.json** file or copy a code that starts `CTC1-`. Campaigns are plain text and cannot run code, so they are safe to pass around. Campaigns live in the browser that made them: keep the file.




**FIG. 7** The campaign library: play, copy, save as a file, copy a share code, import.

WRITING CRISES · CHOOSING TECHNOLOGIES · BALANCE

# CRISES & TECHNOLOGIES

## WRITING A CRISIS

A crisis is a moment that tests one area of both minds. For each one you write what happens, choose the area it tests, and write how each side handles it at each result: SOUND, DELAYED and ERROR. Players read your text at the moment of truth, so keep each line to one or two sentences about a decision, not an outcome.

> **Pick the area by the decision.** A flood of rumours tests NOTICE; a contested video tests INTERPRET; a deadline tests DECIDE; a delivery that must happen tests ACT; a scandal tests TRUST.

The last crisis counts double. Spread the tested areas so no single defence wins the campaign.

## CHECKING BALANCE

**CHECK BALANCE** plays your campaign 40 times, computer against computer, and reports how often each side wins. The built-in campaigns are tuned so Blue wins roughly 40–60% of games. If yours is far outside that, soften the twist or move a crisis to an area the stronger side has not prepared.

**ORIGINAL FICTION**

The built-in campaigns invent every place, outlet, person and event. Doing the same keeps a session about methods rather than about a real event, and keeps it usable in any room.

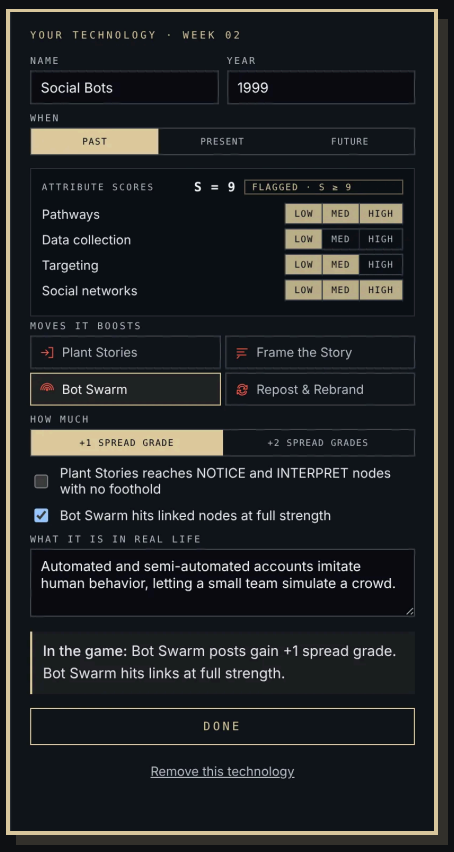



**FIG. 8** Building a technology: name, year, era, the four attribute scores and their total S, and which moves it boosts.

## TECHNOLOGIES

Pick from 18 real technologies, past, present and future, or **BUILD YOUR OWN**. Each is scored 1 to 3 on four attributes from Rushing, Umeokolo & Xu (2025): new pathways, data collection, targeting and social networks. A total of 9 or more is the paper's mark of a disruptive innovation. Where the paper gives no date or score, the editor marks the figure as the designer's estimate.

OFFLINE · PROJECTORS · SOUND · PRIVACY

# SETTING UP THE ROOM

## GETTING IT ONTO MACHINES

| OPTION | BEST WHEN |
|---|---|
| **Play in the browser** (playtest page) | Internet is available and nothing can be installed. |
| **One HTML file** | Locked-down or offline machines: copy it over and double-click. |
| **Installers** (Windows, Mac, Linux) | Personal machines. Not yet code-signed: Windows needs More info → Run anyway; Mac needs right-click → Open the first time. |

## ON A PROJECTOR

Settings → **Chart Room** (the light scheme) reads best on washed-out projectors. Set **Interface size** to LARGER or LARGEST for the back row, and turn on **Colour-safe mode** so the sides differ by shape as well as colour.

## PRIVACY

The game has no accounts, no analytics and no server; its only network request is for web fonts, and it works without them. Settings, saved two-player games and custom campaigns stay in the browser on that machine. Clearing the browser's site data removes them.

## IMMERSIVE BRIEFINGS IN A ROOM

The optional voice pack reads each week in review aloud as the broadcaster of the side ahead (Field Manual page 20). In a room it works best on the presenter's machine with speakers.

1. Download **CognitiveTerrain-Voice-Emma.zip** (about 117 MB) once, before the session.
2. Web or HTML file: Settings → Immersive Briefings → **LOAD VOICE PACK...**. Installers: put the zip beside the game before starting it.
3. Press **HEAR A SAMPLE** to check the speakers.

It runs offline, on that machine only. The voice is synthetic speech from an open-source model; tell your audience so. Slower machines take a few seconds before the first line.

### ACCESSIBILITY CHECKLIST

Text size and interface size are separate settings. Every colour scheme passes 4.5:1 contrast. Reduced motion stops pulsing and drifting. Number keys pick moves; Esc cancels. Captions show whenever the narrator speaks.

DESIGN NOTES · KEVIN WILLIAMSON · 1 OF 4

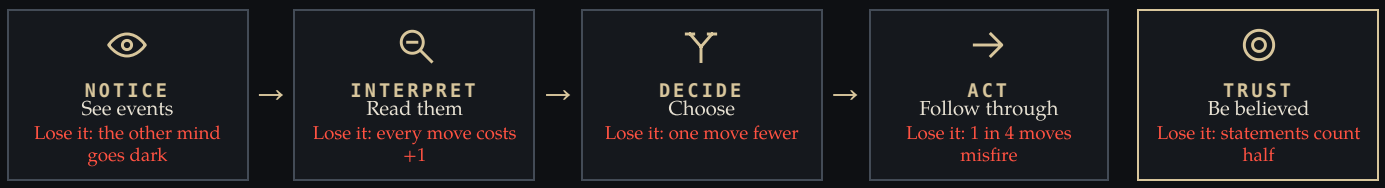
# WHY IT WORKS THIS WAY

*Each mechanic below says what it models, how I arrived at it, and where it comes from. Two kinds of tag show the difference at a glance.*

FROM RESEARCH A published source shaped this rule.

DESIGNER'S ASSUMPTION My own design choice, made so the game plays well.

THE BOARD: FIVE AREAS OF ONE MIND




## THE MIND AS TERRAIN

**MODELS** A nation's decision-making as the ground being fought over.

Two brains of about thirty connected nodes. Every node is a real kind of person or institution: a newsroom, an analysts' cell, a cabinet, a crisis team, the public's trust in its own government.

> **HOW I GOT THERE** *Cognitive warfare is hard to teach because you can't see it, and people picture it as "fake news". I wanted the real target on the table: whether the other side can still decide well under pressure.*

RUSHING, HERSCH & XU (2026)

## LOSING AN AREA

**MODELS** What it costs to lose a capability.

An area falls when the other side holds half its nodes. Each loss has one plain effect (the strip above), so players feel it the way a staff would.

> **HOW I GOT THERE** *The five areas follow the research's cognitive OODA loop with trust underneath. The effects are deliberately simple so they can be read off the map.*

OODA FRAMING: RUSHING ET AL.

EFFECTS: DESIGNER'S ASSUMPTION

## DECISION HEALTH

**MODELS** How well a side can still decide, 0 to 1.

The score is not views or nodes held. It is the health of the four loop areas, multiplied by trust, minus a penalty for each area lost. The math is on page 16.

> **HOW I GOT THERE** *Spreading far is not the same as changing minds. A centralized state leans 40% on DECIDE (Blue 30%): when everything runs through the top, the top is the target.*

RUSHING ET AL. · WEIGHTS AFTER SHIN (NIU)

WEIGHTS: DESIGNER'S READING

## ENTRY POINTS AND FOOTHOLDS

**MODELS** Messages need a way in before they can travel.

Each brain has three entry points: News Media, Social Media and Government Contacts. Attacks start there and spread along links from nodes you have a foothold in.

> **HOW I GOT THERE** *It makes players plan an approach instead of hitting the most valuable node first, which is how influence work actually has to be sequenced.*

DESIGNER'S ASSUMPTION

> **"SPREADING FAR IS NOT THE SAME AS CHANGING MINDS."**

DESIGN NOTES · 2 OF 4 · MOVES, SPREAD AND FAILURE

# MOVES & METRICS

*Eight moves, one engagement metric, and odds that change when a side prepares.*

Plant Stories | Frame the Story | Bot Swarm | Repost & Rebrand | Watch Team | Prebunk | Official Statement | Lessons Learned

## EIGHT MOVES, EIGHT CAPABILITIES

**MODELS** The attacker's and defender's capability families.

Four attack cards and four defence cards, one for each capability family in the research. Military mode keeps doctrinal names. Donovia adds a ninth, Show of Force.

> **HOW I GOT THERE** *I did not invent a list of tricks. Mapping one card to each family keeps the game honest to the framework and easy to cite in a debrief.*

RUSHING, HERSCH & XU (2026) §2.2.2, §2.3.2

SHOW OF FORCE: ISW, IISS

## THE SPREAD SCORE

**MODELS** How engaging a campaign of posts actually was.

Every attack rolls an audience and is graded with the research's engagement metric. The grade then multiplies the move's push:




> **HOW I GOT THERE** *I kept the formula and grade scale because they teach that interaction outweighs views. The multipliers are mine. A viral post on the wrong area still loses the crisis.*

METRIC AND GRADES: RUSHING & XU (2025)

MULTIPLIERS: DESIGNER'S ASSUMPTION

## FAILURE AND BACKFIRE

**MODELS** Platforms, watchers and prepared audiences.

Any attack can be taken down or fall flat. Watching and prebunking raise those odds, and a failed attack nudges the node back toward the defender (page 17 has the numbers).

> **HOW I GOT THERE** *I wanted defence to change the odds, not just add hit points, so players feel why detection and inoculation matter before a campaign lands.*

DESIGNER'S ASSUMPTION

## RECOVERY, FRAMING AND SPREAD

**MODELS** Minds drifting back, and ideas settling in.

Each week every node recovers a little toward its own side; strongly held nodes push on their neighbours. Framing stops recovery for two weeks (three when Donovia frames), so the next push sticks: the game's version of priming.

> **HOW I GOT THERE** *Tuned so that a single hit fades but a sustained, sequenced campaign holds.*

DESIGNER'S ASSUMPTION

> **"DEFENCE SHOULD CHANGE THE ODDS, NOT JUST ADD HIT POINTS."**

# THE MOMENTS THAT COUNT

*Sixteen weeks so it fits a lunch break; four crises so preparation pays.*

| 01 | 02 ◆ TECH | 03 | 04 CRISIS | 05 | 06 ◆ TECH | 07 | 08 CRISIS | 09 | 10 ◆ TECH | 11 | 12 CRISIS | 13 | 14 ◆ TECH | 15 | 16 CRISIS ×2 |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|

## ⚠ FOUR CRISES

**MODELS** Decisions under pressure at the moments that matter.

In weeks 4, 8, 12 and 16 a crisis tests one area of both minds; the last counts double. The result depends on overall health and, heavily, on the tested area (page 16).

> **HOW I GOT THERE** *Weighting the tested area makes players read the calendar and prepare, which is the skill I wanted to reward.*

OODA FRAMING: RUSHING ET AL.

SCORING: DESIGNER'S ASSUMPTION

## ✓ COGNITIVE SUPERIORITY

**MODELS** One side's decision-making clearly dominating.

If one side has lost at least two more areas than the other for two weeks running, the campaign ends early. The race shows in the side panel as LEAD.

> **HOW I GOT THERE** *The research defines cognitive superiority as a sustained advantage in the decision loop; the knockout gives that idea a hard edge.*

RUSHING, HERSCH & XU (2026)

THRESHOLDS: DESIGNER'S ASSUMPTION

## ✳ NEW TECHNOLOGIES

**MODELS** Disruptive innovations arriving mid-campaign.

Both sides choose in secret: use it for an attack bonus, or prepare and deny the other side's. Using it against a prepared opponent gains nothing.

> **HOW I GOT THERE** *A dilemma, because that is the real problem with new tools: you don't know whether the other side is ready. The scoring and the threshold of 9 are the paper's.*

RUSHING, UMEOKOLO & XU (2025)

SOME SCORES: DESIGNER'S ESTIMATE

## $ BUDGET AND MOVES

**MODELS** Finite attention and resources.

Three moves a week from a budget that grows with what you hold, capped at 18, with one extra move for points left over. Access to officials pays more than any post.

> **HOW I GOT THERE** *The cap stops hoarding; the extra move rewards spending.*

DESIGNER'S ASSUMPTION

> **"READ THE CALENDAR, AND PREPARE THE AREA THE NEXT CRISIS TESTS."**

# RED IS NEVER A MIRROR

| | BLUE | OLVANA | DONOVIA |
|---|---|---|---|
| IS | An open democracy | A closed, one-party state | An oligarchic state at permanent information war |
| STRENGTH | Free press, truth that sticks, cheap extra move | Censored internet, state-funded content, already inside | Floods, planted premises, dormant narratives, shows of force |
| WEAKNESS | Its open internet lets Red's posts spread | Runs from the top; slow to learn | Predictable; threats can blow back |
| SHOULD FEEL | Under pressure, keeping its head | A fortress with one gate | Loud |

## OPEN AGAINST CLOSED

**MODELS** Why open and closed societies are vulnerable in different ways.

Blue's posts into a censored internet spread less; Red's into Blue's open internet spread further. Blue gets trust that recovers, statements that hit harder while trust holds, and truth that sticks.

> **HOW I GOT THERE** *The asymmetry is the point: you fight a closed system differently from how it fights you.*

ATP 3-13.1 · SHIN (NIU)

SOME EFFECTS: DESIGNER'S ASSUMPTION

## TWO OPPONENTS, KEPT FICTIONAL

**MODELS** Two doctrines of cognitive warfare.

Red's countries come from the Army's training world, so they are familiar and clearly fictional. Every rule names its source or is labelled as my assumption; every claim was checked against the sources in 0.14.2.

> **HOW I GOT THERE** *Both share a core and differ in temperament, so each feels different with a person playing it.*

ODIN DATE · NIU · EADS ET AL.

NATO C2COE · ISW · IISS · CARNEGIE

## AN OPPONENT THAT PLAYS FAIR

**MODELS** A competent adversary at three levels of skill.

Recruit is noisier and earns 2 fewer points a week. Veteran plans two moves ahead on a scratch copy of the game: no extra budget, no hidden information, never the game's own dice.

> **HOW I GOT THERE** *Very hard to beat, without cheating. Each of its moves also shows the likely reason, judged only from what the player could see.*

DESIGNER'S ASSUMPTION

## BUILT TO BE USED

**MODELS** A teaching game that runs anywhere.

It explains itself at every moment that matters. Two players need only a turn code; campaigns are plain text that cannot run code; Immersive Briefings are optional; every scheme passes 4.5:1 contrast.

> **HOW I GOT THERE** *Keep the original game: new features go around the core loop, never instead of it. A campaign is balanced when Blue wins 40–60% of computer games.*

DESIGNER'S ASSUMPTION

> **"VERY HARD TO BEAT, WITHOUT CHEATING."**

# THE MATH, SHOWN

*Every number the game uses to score a campaign, with a worked example. Nothing here is hidden from players; the side panel and the week in review show the same sums.*

## 1 · AREA HEALTH

```
area = average over its nodes of
clamp( (lean + 50) ÷ 150, 0, 1 )
```

A node's lean runs from −100 (held hard by the enemy) to +100 (held hard by its owner). A node fully yours counts 1; a neutral one about 0.33; one the enemy holds at −50 or worse counts 0. Partial credit keeps the score from lurching.

## 2 · DECISION HEALTH

```
areas = w·NOTICE + w·INTERPRET + w·DECIDE + w·ACT
DQ = areas × ( 0.75 + 0.25 · TRUST )
     − 0.06 × areas lost − flood
```

| WEIGHT | NOTICE | INTERPRET | DECIDE | ACT |
|---|---|---|---|---|
| BLUE | 0.20 | 0.25 | 0.30 | 0.25 |
| OLVANA · DONOVIA | 0.15 | 0.20 | 0.40 | 0.25 |

Trust scales everything: with trust gone, the most a side can score is 0.75. **Flood** (Donovia only): −0.02 for each of its attacks that lands in a week beyond the first two (one more is free per Lessons Learned level), at most −0.06, until Donovia's next turn.

### WHAT MOVES THE NEEDLE

| CHANGE | DECISION HEALTH | NEXT CRISIS |
|---|---|---|
| Tested area +0.10 | +0.01 to +0.04 | **+0.05 to +0.06** |
| Any other area +0.10 | +0.01 to +0.04 | +0.01 to +0.02 |
| TRUST +0.10 | about +0.02 | about +0.01 |
| Lose an area | −0.06 and more | −0.04; −0.12 if tested |

The lesson in numbers: the area the next crisis tests is worth about three times any other. That is why Looking ahead names it.

## 3 · A CRISIS

```
crisis = 0.6 · DQ + 0.4 · tested area
− 0.08 if the tested area is lost
```

| ERROR | DELAYED | SOUND |
|---|---|---|
| below 0.72 · 0 pts | 0.72+ · 1 pt | 0.86+ · 3 pts |

**Plus 1** to a side whenever the other side scores ERROR. Week 16 doubles everything, bonus included.

The "decides in … hours" shown in the game is 6 ÷ score², rounded.

### WORKED EXAMPLE

Blue's areas: NOTICE 0.9, INTERPRET 0.8, DECIDE 1.0, ACT 0.9; TRUST 0.7; nothing lost.

```
areas = .2×.9 + .25×.8 + .3×1 +
.25×.9 = 0.905
trust = 0.75 + 0.25×0.7 = 0.925
DQ = 0.905 × 0.925 = 0.84
```

A crisis testing INTERPRET: 0.6×0.84 + 0.4×0.8 = **0.82**, so **DELAYED**, 1 point. Lift INTERPRET to 0.9 and DQ rises too: about **0.88**, **SOUND**, 3 points.

## 4 · WINNING

Most crisis points after week 16 wins. **Tied?** The side whose Decision Health is more than 0.03 higher wins; otherwise it is a draw. **Knockout** earlier: the other side has lost at least 2 areas, and at least 2 more than you, at the end of 2 weeks running.

# ODDS & BALANCE

*How hard a move hits, how often it fails, what comes in each week, and how far to trust a balance check.*

## 5 · SPREAD SCORE AND GRADE

```
score = ( 0.1·views + 0.3·likes +
0.7·comments + 1·shares ) ÷ posts
```

| GRADE | SCORE | PUSH × |
|---|---|---|
| F · E · D | 0–2 · 3 · 4–9 | 0.25 · 0.45 · 0.7 |
| C · B | 10+ · 100+ | 1.0 · 1.2 |
| A · A+ | 1,000+ · 10,000+ | 1.4 · 1.6 |

Reach is rolled on a ten-times scale, so ×10 is about one grade. **About a grade up:** their TRUST lost; each step of a technology you used. **About a grade down:** a Watch Team on the target. **Smaller nudges:** an entry point ×2.5; holding their News Media ×3; Red posting into Blue's open internet ×1.25; Blue posting into Olvana's censored internet ÷1.4 (Donovia's ÷1.25).

## 6 · PUSH

```
push = card power × area fit × grade
× 0.5 if prebunked (0.45 for Donovia's
attacks)
× 1.4 if framed × 2 if it wakes a dormant
narrative
```

Card power: Plant Stories 20 (×1.5 on an entry point), Frame 16, Bot Swarm 12 plus 7 on every link, Repost 10, Show of Force 16. **Area fit** runs 0.75 to 1.5: a card's best area (Field Manual Quick Reference) gets 1.5; Repost & Rebrand is 1 everywhere.

> **Example.** Plant Stories on their News Media (an entry point, in NOTICE, the card's best area), graded B: 20 × 1.5 × 1.5 × 1.2 = **54**. Lean runs −100 to +100: a node is **held** at 30 toward a side and gives a **foothold** at 10.

## 7 · ODDS OF FAILURE

| ATTACK FAILS IF… | CHANCE |
|---|---|
| Any attack | 8% |
| + target watched | +15% |
| + target prebunked | +8% |
| + Blue posting into Olvana / Donovia | +2% / +1% |
| Show of Force | +4% |
| Official Statement ignored | 6% (+20% if TRUST lost) |

A failed attack pushes the node back 6 toward the defender (9 when Donovia's fail); a failed Show of Force also pushes each neighbour back 4. With ACT lost, 1 move in 4 misfires before any of this is rolled.

## 8 · EACH WEEK

**Budget:** 6 + 1 per enemy node held (Top Leadership 2) + 2 for their Government Contacts; at least 2, at most 18. Recruit Red −2; Donovia +1. Extra move: Blue 2, Red 5.

**Recovery:** each node +1 toward its owner (+1 per Lessons Learned level, half for Red), halved with TRUST lost. Blue's TRUST nodes +2 more; Red's entry points ×1.2 (Donovia's ×1.1); Red's nodes leaning Blue only ×0.35. None while framed or once captured. Any neighbour leaning 50+ the other way pushes 5 (half if prebunked).

## 9 · READING A BALANCE CHECK

The target is Blue winning 40–60% of computer games. **CHECK BALANCE** plays 40, so a result is only good to about ±15 points: 35% might really be 50%. Treat 30–70% as "probably fine", and run it twice before changing anything.

| BLUE WON (OF 40) | LIKELY TRUE RANGE |
|---|---|
| 12 · 30% | 16–44% |
| 20 · 50% | 35–65% |
| 28 · 70% | 56–84% |

95% ranges. For a firmer answer, run CHECK BALANCE several times and add up the wins.

# WHERE THIS COMES FROM

*The game is built on published research. The rules simplify it; the debrief and the in-game guide point back to it.*



## REFERENCES


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### ATTRIBUTION AND LICENSE

This game and manual adapt concepts, the capability taxonomy, the engagement-effectiveness metric (formula, interaction weights and grade scale) and two sets of innovation attribute scores from the three Rushing et al. papers above. Each paper is licensed under Creative Commons Attribution 4.0 International (CC BY 4.0): creativecommons.org/licenses/by/4.0. Papers: arxiv.org/abs/2603.05222 · arxiv.org/abs/2510.15805 · arxiv.org/abs/2510.15801.

**Changes made:** the material was adapted into game mechanics and rewritten for play. Engagement grades are mapped to effect multipliers invented for this game; the OODA effects, capability families and cognitive superiority definition are expressed as game rules; three of the five innovation score sets are the designer's own estimates; all scenario, narrative and tutorial text is original.

**No endorsement:** the authors and their institutions have not reviewed or endorsed this game.

**Immersive Briefings voice pack** (optional, separate download): Kokoro-82M by hexgrad, kokoro-js and transformers.js (Apache 2.0); ONNX Runtime (MIT); eSpeak NG (GPL-3.0-or-later), whose source is published with the pack.

### HOW THIS GAME WAS MADE

**Design and direction:** Kevin Williamson. Cognitive Terrain is my design: I conceived the game, built its rules from the research, chose every source, defined both opponents and the campaign briefs, set the balance targets and accessibility standards, and reviewed every rule, campaign and page.

**Tools:** I used AI-assisted development tools to help implement the code, draft text and run balance tests, working from my specifications; I reviewed, revised or rejected everything they produced. The published sources, not the tools, are the authority for any rule that models real-world behaviour. The art, icons and music are drawn and synthesized by the game's own code; no AI image or music generators were used, and the base game runs no AI while you play. The optional Immersive Briefings voice is synthetic speech from an open-source model that runs on your own computer (Field Manual page 20).

Olvana and Donovia, and the countries Bothnia, Framland, Arnland and Gorgas, are fictional training countries from the US Army's ODIN DATE environment; they are used here without endorsement. All eight scenarios, their towns, outlets, people and every event in them are original fiction. Any resemblance to real places or events is coincidental. Donovia's rules follow the NATO, ISW, IISS and Carnegie research above; game effects are abstractions of broad, publicly reported patterns, not claims about any specific operation.